
PROTOCOL

1. **Photograph the plates twice** — under **blue-light transillumination** and under **ambient light**. Save both. Once you have picked, these are the only record of what the colonies looked like.
2. **Tubes or a block**. Up to **4 cultures**: snap-cap tubes, each labelled with its clone ID. More than that: a **24-well block**, labels on the **airpore sheet** — the wells are identified by position.
3. Fill each tube or well with **4 mL 2YT + antibiotic** — the one on your labsheet. **It is not always carb.**
4. Pick **4 colonies from each plate**: touch a sterile toothpick to a single colony and drop it in. It stays there.
 - **Pick the ones that are easy to pick and look healthy** — well separated, round and smooth. Big and well isolated is usually best.
 - **Avoid contorted or smudgy colonies**. That look means water has seeped across the plate, or two colonies have grown into one.
 - **Which colonies to choose is set by the experiment** — brightest, smallest, largest, red, or deliberately naive. **Your labsheet says which.**
 - **Into a block, follow the grid on your labsheet**. Wells are identified by position, so a block filled in a different order is silently scrambled — nothing errors. Labelled tubes do not have this problem.
5. **Cover**. Tubes: caps on loosely, so they can breathe. Block: the airpore sheet.
6. Grow in the **shaking incubator, 16–36 h**. Into the fridge by **36 h** — past that they overgrow.
7. **Parafilm the plates** and store them **upside-down in the fridge** — you may need to go back to them.